

FactGrid Cuneiform Discovery Project

- FactGrid AWCA Google Drive & Google Colab
- FactGrid Cuneiform AWCA GitHub Org. Repos

Circle Chen	2025
Daisy Wang	2025
Zaid Maayah	2024
Floyd Fang	2022
Tina Chen	2024
Kevin Gao	2023
Conner Mi	2023

Discovery Students 2021-2022

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Minoo Kim	2022
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The goal for this project is, first, to link every cuneiform artifact on record to the primary and secondary sources in publicaition. Once this is accomplished we can extend this referencial system to include the entities named on each artifact (i.e. people, places, and things)

This project is inspired by the durability of the data preserved in the oldest writing system known to mankind, called **cuneiform**. There are approximately a half-million artifacts with cuneiform writing spread all over the planet. Many of these objects are not even photographed, let alone translated. Scholars in this field have made a number of relational text databases, in order to identify these objects housed in museums and private collections, and while these databases have helped create a system of identificaiton and textual analysis, they have yet to be **linked** together to each other and to the existing scholarship.

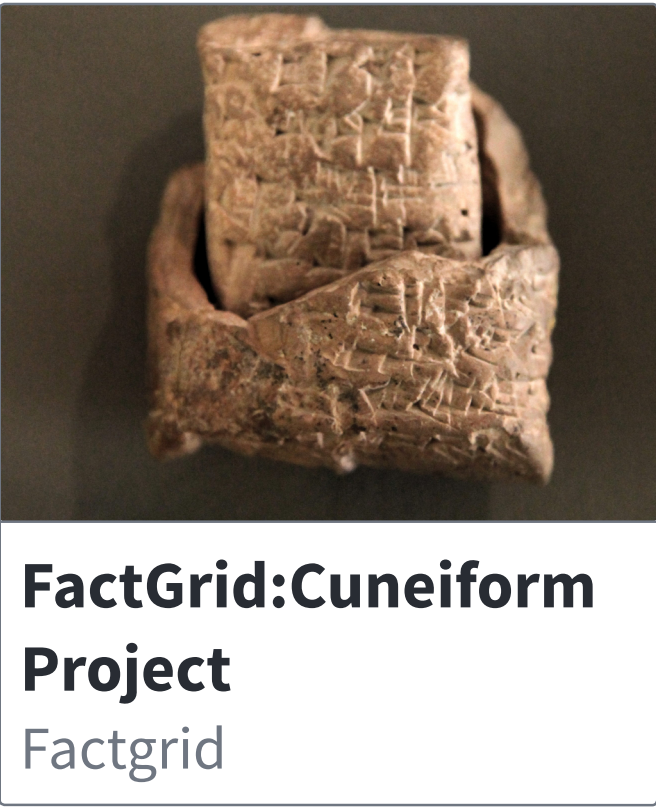
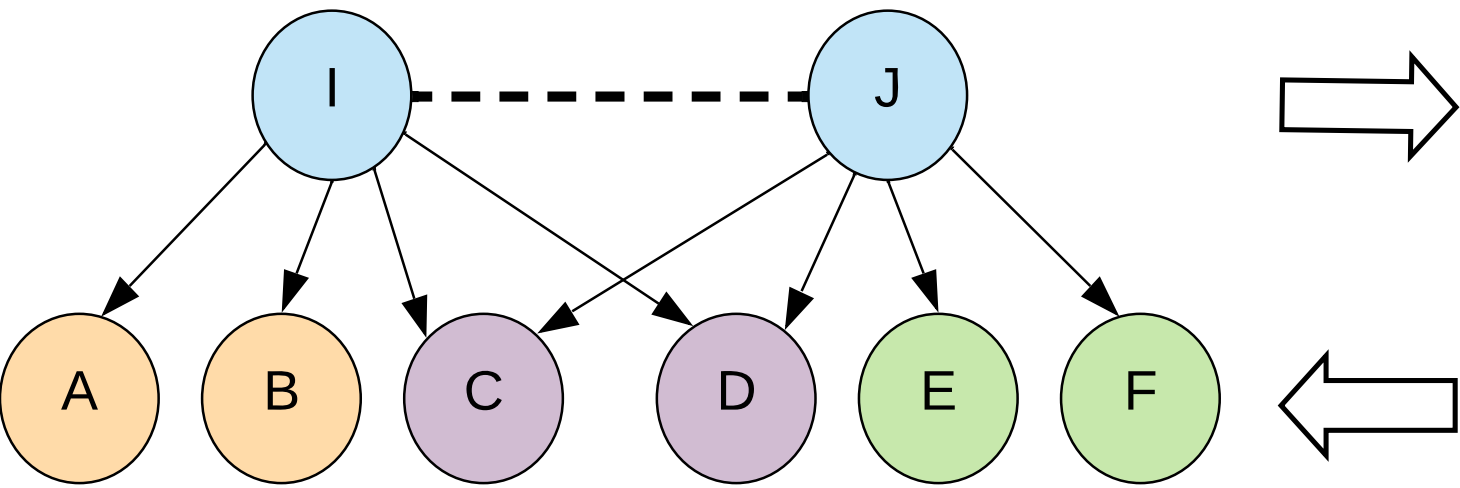
Our Cuneiform project in FactGrid is building language support for all languages written in cuneiform, a writing system used for about 4000 years (from 3200 BCE to 50 CE). We are working with more than 350k documents to build dictionaries for these languages and social network graphs for the people, places, other entities named in these texts.

The main challenge we're working on is how to build reproducible workflows for linking four online databases of cuneiform sources (each with more than 100k documents) with two datasets of secondary sources. We're using Python notebooks (ipynb) to harmonize these open source databases, and we are linking the results using FactGrid Cuneiform, which is a triple store (or database for RDF triple statements). Our work helps us deepen our knowledge of Python for NLP and SparQL, the query language for Wikidata.

FactGrid is a Wikibase triplestore designed for historical research, which makes this the ideal hub for linking the existing scholarship, both primary and secondary sources, for every cuneiform artifact in publication. As a graph database (using RDF triples), this system allows for an expansive number of relationships to exist between any objects or entities, and for these relationships to be classified for greater discoverability. Additionally, the structure of a graph database provides simple solutions to duplication, disambiguation, and alternative editions of the same entity, which allows for a more comprehensive approach for linking the different datasets in various languages. Because the database is in Wiki(data) format, this provides an open forum for editing and collaborating internationally, which makes the data more robust, timely, and sustainable.

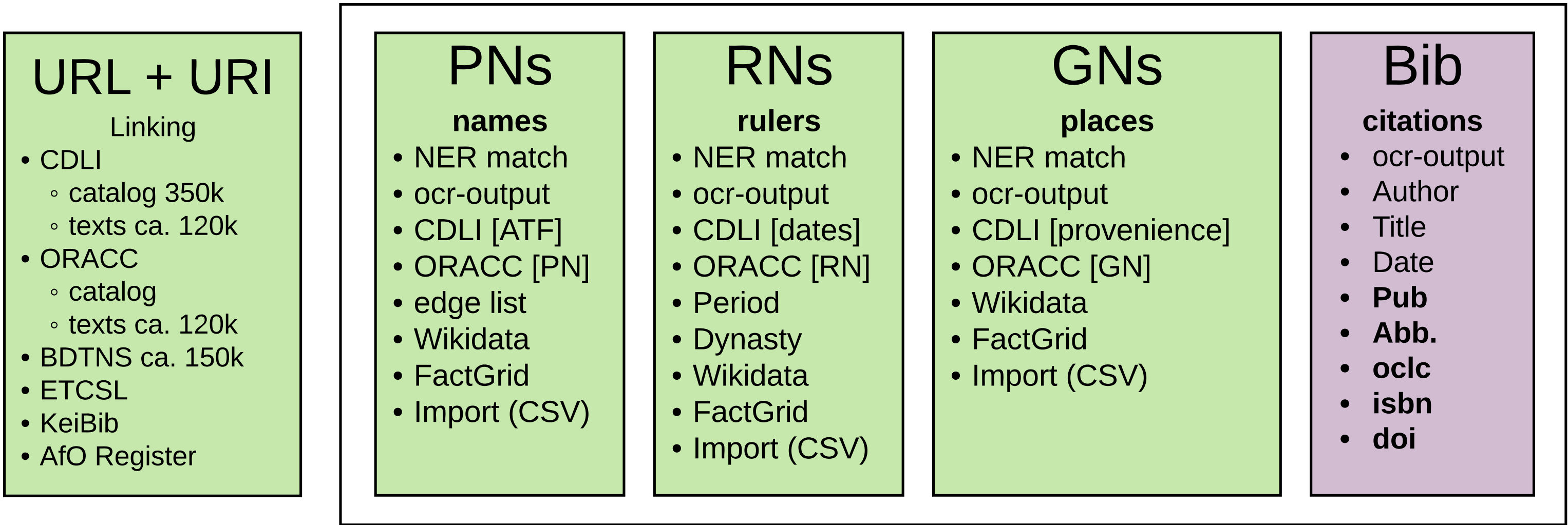
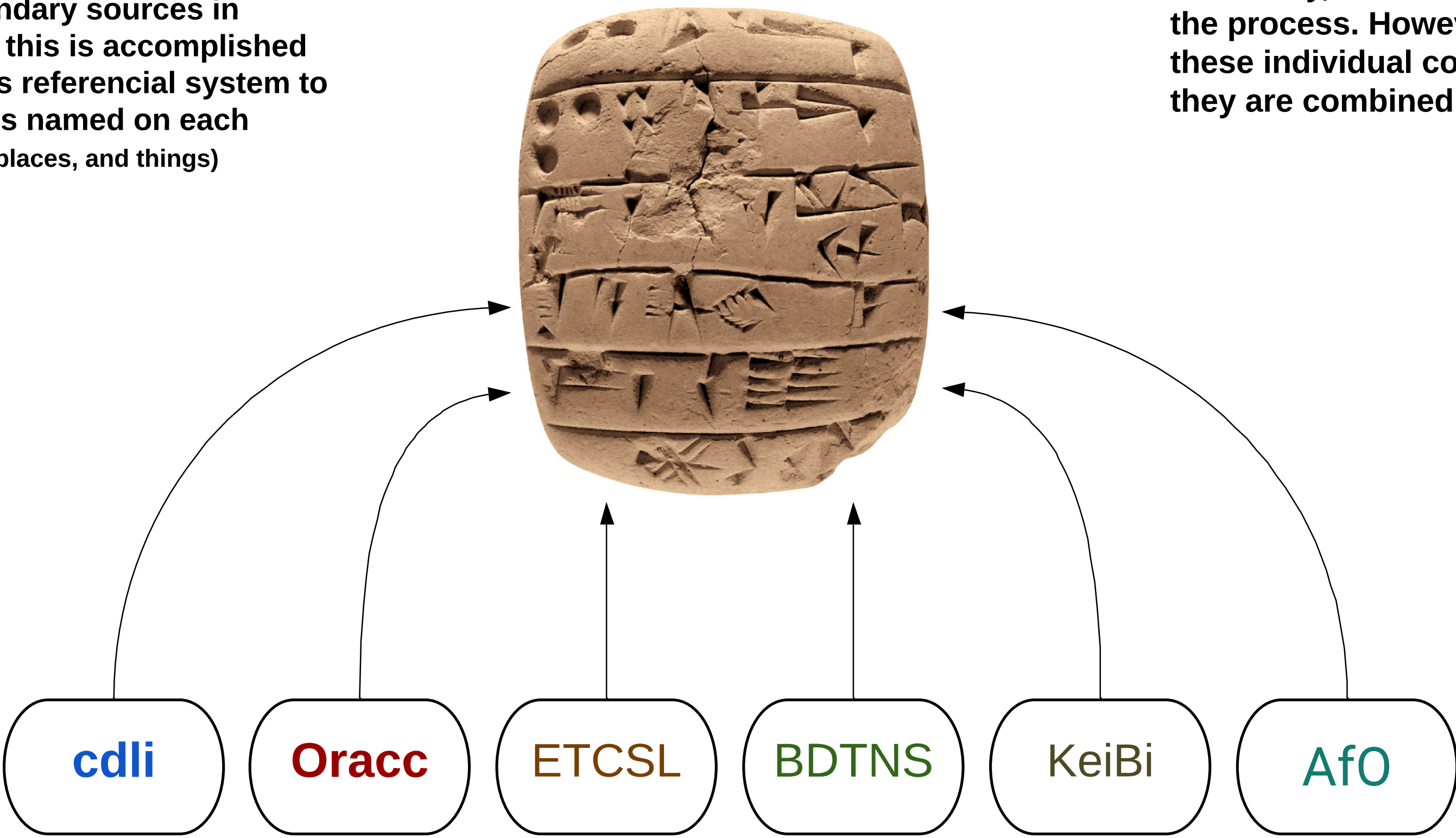
Concepts of Linked Data and bibliographic coupling

Entities I and J are items with URIs, for example in Wikidata. Such items can be attested bibliographically and cited by documents C and D. The documents which cite these entities can be bibliographically coupled and their relational values can be measured by a count-frequency and cosine similarity scores.



"Data is a precious thing and will last longer than the systems themselves."

Tim Berners-Lee, Inventor of the World Wide Web



Cuneiform Digital Library Initiative

Improving the Sumerian language linguistic annotation pipeline

- cdli-gh GitHub Org. Repos

The goal of this project is to improve the linguistic (manual and automatic) annotation pipeline for the Sumerian language. We have developed a large range of state-of-the-art tools, dictionary, rule and ML based for each step of the process. However, we can improve most of these individual components and also the way they are combined, and the workflows involved.

Discovery Students 2022

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Barney Zhongyi Tu Li	2025
Mei Li Johnson	2023
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The Cuneiform Digital Library Initiative (CDLI <https://cdli.mpiwg-berlin.mpg.de/>) is a worldwide group of scholars and scientists who digitize, curate, analyze, and share ancient cuneiform inscriptions dating from the beginning of writing. One of the main challenges of CDLI is to find ways to render these ancient texts more accessible to a wider range of audiences. Most remain unannotated and untranslated and thus impenetrable to the uninitiated. During the term, students worked on improving the linguistic annotation pipeline for cuneiform texts in the Sumerian language, converting annotations to open and sharable formats, and lexicalizing the initiative, by preparing the way to harvest word tokens for the internal glossary at the initiative. Efforts have resulted in:

- A linguistically annotated corpus was processed, cleaned and inserted in the database for easier reuse
- An important CoNLL format converter was improved so texts could be stored in the proper format for reuse
- Efforts were deployed in preparing a harvesting workflow for word tokens and lemmata
- Efforts were deployed in the design of a parser to check text validity

These contributions all fit into the larger puzzle of addressing this access challenge and because of this work, we are one step further toward connecting the efforts deployed at the CDLI and those in the Cuneiform Wikidata group.

CDLI-CoNLL-to-CoNLLU-Converter

Cleaning Conll Output of the Royal Subcorpus

ID	FORM	SEGM	XPOSTAG	HEAD	DEPREL	MISC
a.1.1	da-da	da-da[3]	PN	11	—	—
a.2.1	ensi2	ensi2[ruler]	N	1	—	—
a.3.1	szuruppak{ki}	szuruppak{ki}[1]	SN.GEN	2	—	—
a.4.1	ha-la-ad-da	ha-la-ad-da[1]	PN	1	—	—
a.5.1	ensi2	ensi2[ruler]	N	1	—	—
a.6.1	szuruppak{ki}	szuruppak{ki}[1]	SN.GEN	5	—	—
a.7.1	dumu-ni	dumu[child]	N.3-SG-H-POSS.ERG	1	—	—
a.8.1	ad-us2	ad-us2[plank]	N.ABS	11	—	—
a.8.2	abul	abul[gate]	N	11	—	—
a.9.1	{d}sud3-da-ke4	{d}sud3[1]	DN.GEN.L3-NH	9	—	—
a.1.0.1	bi2-in-us2	us2[follow]	3-NH.L3.3-SG-H-A.V.3-SG-P	0	—	—

FactGrid Cuneiform & Wikidata Lexemes

